

Motion — Kinematics

📁 1.2 Motion 🌐 IGCSE Physics 0625 – Extended 📄 Version 1 📅 2026-06-30

History

The study of motion, known as **kinematics**, began with ancient Greek philosophers.

Aristotle (384–322 BCE) believed that objects fell at speeds proportional to their weight — a heavier object falls faster than a lighter one. This idea was accepted for nearly 2,000 years because it seemed intuitive.

Galileo Galilei (1564–1642) challenged Aristotle's view. Through careful experiments using inclined planes and water clocks, he demonstrated that in the absence of air resistance, **all objects fall at the same rate regardless of mass**. Galileo's work laid the foundation for modern kinematics.

Myth vs Fact: The story of Galileo dropping balls from the Leaning Tower of Pisa is a popular myth. In reality, he used inclined planes to slow down the motion so he could measure it more accurately with the primitive timing tools available.

Sir Isaac Newton (1643–1727) later formalised the laws of motion in his work *Philosophiæ Naturalis Principia Mathematica*, unifying terrestrial and celestial motion under the same mathematical framework.

Nature & Applications

Kinematics — the description of motion — is everywhere in our daily lives:

- 🚗 **Cars & Transport:** Speedometers measure instantaneous speed. Sat-navs calculate average speed and time of arrival.
- 🏃 **Sports:** Athletes analyse split times, acceleration off the starting block, and peak speed.
- 🚀 **Space Exploration:** Rocket launches require precise calculations of velocity and acceleration to reach orbit.
- 🚦 **Safety:** Thinking distance and braking distance determine stopping distances for road safety.

Key idea: Motion is **relative** — we describe it from a chosen reference point (the observer's position).

Theory

Key Quantities

- **Distance (s)** — total path length travelled (scalar, SI unit: metre, m)
- **Displacement (s)** — straight-line distance from start to finish in a given direction (vector, m)
- **Speed (v)** — rate of change of distance: (scalar, m/s)
- **Velocity (v)** — rate of change of displacement: (vector, m/s)
- **Acceleration (a)** — rate of change of velocity: (vector, m/s²)

Equations for Uniform Acceleration (SUVAT)

These equations apply only when acceleration is **constant**:

Where:

- = initial velocity (m/s)
- = final velocity (m/s)
- = acceleration (m/s²)
- = time (s)
- = displacement (m)

Free Fall

When an object falls freely under gravity, **air resistance is negligible**, and the acceleration is constant at:

The equations of motion become:

For a freely falling object dropped from rest ():

Graphs of Motion

Graph Type	Y-axis	Gradient	Area Under
Distance–Time	distance (m)	speed (m/s)	—
Speed–Time	speed (m/s)	acceleration (m/s ²)	distance (m)
Acceleration–Time	acceleration (m/s ²)	—	change in velocity (m/s)

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

$$g = 9.8 \text{ m/s}^2 \text{ (downward on Earth)}$$

$$v = u + gt$$

$$s = ut + \frac{1}{2}gt^2$$

$$v = gt$$

$$s = \frac{1}{2}gt^2$$

$$v = \frac{\Delta s}{\Delta t} \quad v = \frac{\Delta s}{\Delta t} \quad a = \frac{\Delta v}{\Delta t} \quad u \quad v \quad a \quad t \quad s \quad u = 0$$



Horizontal Motion

Experiment with different modes of motion — from rest, constant speed, acceleration, and deceleration.

Free Fall (Vertical Motion)

Observe the motion of a freely falling apple under different gravitational fields.

Kinematics Graphs of Horizontal Motion

v1.2.0

Theme

☒ Distance-Time ☐ Speed-Time ☐ Acceleration-Time

Init Position 0 Init Speed 8.0

DISTANCE-TIME — → CONSTANT SPEED

Time / s	Distance / m
0	0
2	17.6
4	35.2
6	52.8
8	70.4
10	88.0



Kinematics Graphs of Horizontal Motion

v1.2.0

Theme



Distance-Time



Speed-Time



Acceleration-Time

At Rest

→ Constant Speed

↗ Increasing Speed

↘ Decreasing Speed

↗ Constant Acceleration

↘ Constant Deceleration

Init Position



0

Init Speed



8.0



Random

Reset

Play

DISTANCE-TIME — → CONSTANT SPEED

88.0

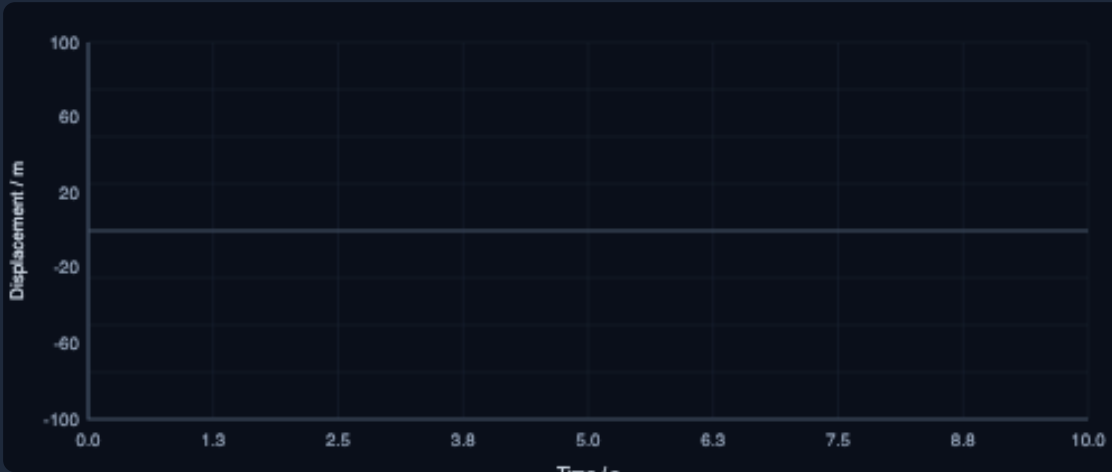


Kinematics of a Freely Falling Body

v1.0.0

Theme

DISPLACEMENT-TIME



VELOCITY-TIME

50



Kinematics of a Freely Falling Body

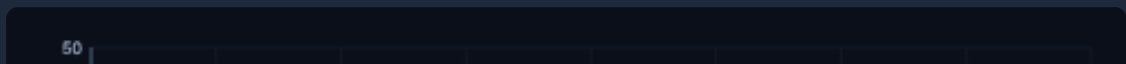
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Theme

DISPLACEMENT-TIME



VELOCITY-TIME



Worked Examples

Example 1: Constant Acceleration

A car starts from rest and accelerates uniformly at for 10 seconds.

(a) What is its final velocity?

(b) How far does it travel?

Solution (a):

Solution (b):

Example 2: Free Fall

A ball is dropped from a height of 45 m. How long does it take to reach the ground? (Take)

Solution:

Example 3: Thinking and Braking Distance

A car travels at . The driver's reaction time is . The braking deceleration is .

(a) What is the thinking distance?

(b) What is the braking distance?

(c) What is the total stopping distance?

Solution (a):

Thinking distance = speed $\times$ reaction time

Solution (b):

Braking distance:

Solution (c):

Total stopping distance =

$$v = u + at = 0 + (2.0)(10) = 20 \text{ m/s}$$

$$s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2}(2.0)(10)^2 = 100 \text{ m}$$

$$s = \frac{1}{2}gt^2$$

$$45 = \frac{1}{2}(10)t^2$$

$$t^2 = 9$$

$$t = 3.0 \text{ s}$$

$$= (20)(0.5) = 10 \text{ m}$$

$$v^2 = u^2 + 2as$$

$$0 = (20)^2 + 2(-4.0)s$$

$$s = \frac{400}{8} = 50 \text{ m}$$

$$2.0 \text{ m/s}^2 \quad g = 10 \text{ m/s}^2 \quad 20 \text{ m/s} \quad 0.5 \text{ s} \quad 4.0 \text{ m/s}^2 \quad 10 + 50 = 60 \text{ m}$$